

JEN-HSIANG OU

I'm a postdoctoral researcher at the Department of Medical Genetics, National Taiwan University Hospital. My research focuses on structural variations, like repeat sequences, and DNA methylation in relation to diseases. I work with both next-generation and long-read sequencing technologies to study the genetic basis of rare diseases. Having completed my doctoral studies at Uppsala University and earned a master's degree in Biostatistics from National Taiwan University, my goal is to better understand genetic variation and its role in disease, contributing to advancements in medical research.

PERSONAL INFORMATION

歐任翔 Jen-Hsiang (Mark) Ou, Ph.D.

Position: Postdoc researcher, Department of Medical Genetics, National Taiwan University Hospital

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EDUCATION

PhD candidate in Computational Genetics 2020-2024

- Department of Medical Biochemistry and Microbiology, Faculty of Medicine, Uppsala University, Sweden
- Thesis: Exploring the Genetic Landscape of Chicken Populations: Admixture, Growth QTLs, and Long-Term Selection Dynamics

Uncover the domestication history of global chicken, the complex genetic architecture of chicken body weight, and the selection response after 40 generations of intense bi-directional selection.

MSc in Biostatistics 2016-2018

- Department of Agronomy, College of Bioresources and Agriculture, National Taiwan University, Taiwan
- Thesis: Training Set Determination for Genomic Selection (doi:10.1007/s00122-019-03387-0)
Provide a new optimality criterion to determine a training set that is expected to result in the highest Pearson's correlation between the genomic estimated breeding value and the actual phenotype.

BSc in Agronomy 2012-2016

- Department of Agronomy, College of Bioresources and Agriculture, National Taiwan University, Taiwan

EXPERIENCES & ACHIEVEMENTS

Postdoc Researcher 2024-recent

- Department of Medical Genetics, National Taiwan University Hospital (NTUH)
- Understanding the genetic basis of the inherited muscular dystrophy disease, Facioscapulohumeral dystrophy (FSHD).

ISAG Conference bursary winner 2023

- International Society for Animal Genetics Conference (ISAG2023)

Organizer and Host of the Department day 2023

- IMBIM-day 2023. A full-day scientific workshop and banquet

TA of Comparative Genomics for Biomedicine Course 2020-2024

- Master-level course at Uppsala University

Teacher of Bioinformatic Course 2020-2024

- Master-level course at Uppsala University
- Responsible for the GWAS module, giving main lectures, serving as an examiner, and practicing TA

Research assistant 2019

- Uppsala Biomedical Centre, Uppsala University, Sweden
- NGS data alignment, quality control, server maintenance, teaching, and software development

Research assistant 2018-2019

- National Taiwan University, Taiwan
- Software developed for simulation studies, server maintenance, and manuscript writing

TA & 3 times excellent teaching assistants 2016-2018

- National Taiwan University, Taiwan

- TA for statistics labs. Teaching the R programming language and practical application of statistics

LANGUAGES & SKILLS

- Mandarin (Native)
- English (Fluent)
- R (Advanced)
- LaTeX/Debian (Advanced)
- Linux (Advanced)
- Bioinformatic tools (GATK, Plink, vcftools, etc.)
- Python (Intermediate)
- C++ (Elementary)
- HTML + CSS (Elementary)

PUBLICATIONS

- Ou, J.H. (2024). Exploring the genetic landscape of chicken populations: Admixture, growth QTLs, and long-term selection dynamics. *Doctoral thesis, Uppsala dissertations from the Faculty of Medicine 2053*.
- Ou, J.H., Rönneburg, T., Carlborg, Ö., Honaker, C.F., Siegel, P.B., Rubin, C.J. (2024). Complex genetic architecture of the chicken *Growth1* QTL region. *PLoS ONE*, 19(5):e0295109.
- Ou, J.H. (2024). geno2r: Functions for reading genotype data in R. *R package version 1.6.2*. <https://www.oumark.net/geno2r/>.
- Ou, J.H., Wu, P.Y., Liao, C.T., (2023). TSDFGS: Training set determination for genomic selection. *R package version 2.4.2*. <https://www.oumark.net/TSDFGS>.
- Wu, P.Y., Ou, J.H., Liao, C.T. (2023). Sample size determination for training set optimization in genomic prediction. *Theoretical and Applied Genetics*, 136(3).
- Rönneburg, T., Ou, J.H., Pettersson, M., Honaker, C.F., Siegel, P.B., Caroborg, Ö. (2023). Within-line segregation as contributors to long-term, single-trait selection-responses in the Virginia chicken lines. *Manuscript in thesis, Uppsala University*.
- Guo, Y., Ou, J.H., Zan, Y., Wang, Y., Li, H., Zhu, C., Chen, K., Zhou, X., Hu, X., Caroborg, Ö. (2022). Researching on the fine structure and admixture of the worldwide chicken population reveal connections between populations and important events in breeding history. *Evolutionary Applications*, 15(4).
- Ou, J.H., Liao, C.T. (2019). Training set determination for genomic selection. *Theoretical and Applied Genetics*, 132(10).
- Lin, P.C., Tsai, Y.C., Hsu, S.K., Ou, J.H., Liao, C.T., Tung, C.W. (2018). Identification of nature variants affecting chlorophyll content dynamics during rice seedling development. *Plant breeding*, 137(3).